Explain local and regional processing and write their steps of operation.

Local Processing

Local processing involves analyzing the characteristics of pixels in a small neighborhood (Sry) around every point that has been identified as an edge point.

Steps of Operation:

1. **Gradient Calculation:** Compute the gradient magnitude $M(x, y)$ and angle $A(x, y)$ for the input image.
2. **Neighborhood Analysis:** For every edge point (x, y), examine all pixels (s, t) in a small predefined neighborhood. +1
3. **Similarity Check:** Link points (s, t) to (x, y) if they satisfy two similarity criteria:
    - **Magnitude Similarity:** The difference in gradient magnitudes is within a threshold E: $|M(x, y) - M(s, t)|[cite_start] \leq E$.
    - **Angle Similarity:** The difference in gradient directions is within a threshold A: $|A(x, y) - A(s, t)|[cite_start] \leq A$.
4. **Edge Formation:** Linked points form a continuous edge segment.

Regional Processing

Regional processing is used when the location of regions of interest is known or can be determined. It often employs polygonal approximations to capture essential shape features while simplifying the boundary representation.

Steps of Regional Processing:

1. **Point Sequencing:** Start with an ordered sequence P of distinct, 1-valued points representing the boundary of a binary image.
2. **Initialize Segment:** Specify two starting points (vertices), A and B, to form an initial line segment.
3. **Distance Calculation:** Calculate the distance of every point in the sequence between A and B from the line segment AB.
4. **Thresholding (D_{max}):** Find the point V_{max} with the maximum distance D_{max} from the segment.
5. **Vertex Addition:** If $D_{max} > T$ (a predefined threshold), V_{max} is declared a new vertex.
6. **Recursion:** Repeat the process recursively for the new segments (e.g., AV_{max} and $V_{max}B$) until all points on the original boundary are within distance T of the nearest line segment.